

TRADEPASS ACADEMY

309A COMPANION SERIES · VOLUME 1

Formula Quick Reference

Every formula tested on the Construction Electrician 309A exam, with worked examples and quick-reference tables.

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- Ohm's law, power, energy and AC fundamentals
 - Single-phase and three-phase power formulas
 - Transformer, motor and voltage-drop calculations
 - Unit conversions, constants and electrical symbols
 - Worked examples in exam format throughout

Companion resource to: Red Seal Construction Electrician 309A Exam Prep, 2026–2027 Edition

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How to use this reference

This quick reference gathers every formula you are expected to command on the Red Seal Construction Electrician 309A examination. It is a companion to your TradePass Academy study guide, not a replacement for it: the book teaches each concept, and this reference is what you drill from once the concept is understood.

The exam is closed-book for formulas. Work through each section until you can write the formula from memory, then confirm your recall against this sheet. In the final week, use only the formula index on the last page and reconstruct everything else from memory.

EXAM TIP

Formulas are rarely tested naked. The exam wraps them in a job situation: a heater on a 240 V circuit, a motor at the end of a long feeder. Practise translating word problems into knowns and unknowns before you reach for the formula.

1. Electrical constants and standard values

Commit these constants and Canadian standard values to memory. They appear inside calculation questions without being stated.

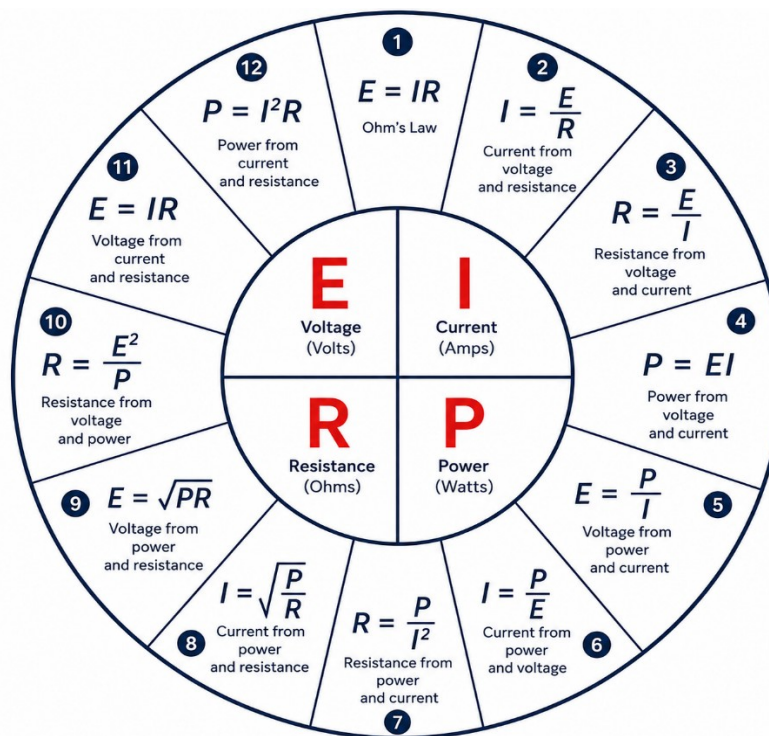
Constant / value	Figure	Used for
Resistivity of copper (ρ , at 20 °C)	$1.72 \times 10^{-8} \Omega \cdot m$	Conductor resistance, voltage drop
Resistivity of aluminum (ρ , at 20 °C)	$2.83 \times 10^{-8} \Omega \cdot m$	Conductor resistance, voltage drop
K factor, copper (≈ 75 °C)	$\approx 12.9 \Omega \cdot \text{cmil}/\text{ft}$	Imperial voltage-drop method
K factor, aluminum (≈ 75 °C)	$\approx 21.2 \Omega \cdot \text{cmil}/\text{ft}$	Imperial voltage-drop method
1 horsepower	746 W	Motor calculations
Canadian utilization frequency	60 Hz	Reactance, motor speed
$\sqrt{3}$ (three-phase factor)	1.732	All three-phase power work
Common 1 ϕ system voltages	120/240 V	Dwellings, small services
Common 3 ϕ system voltages	120/208 V, 347/600 V	Commercial and industrial
Water heats / standard temp rise	1 kWh raises $\approx$ heat questions	Rare; read the question data

WATCH OUT

Canadian exams use metric-first values with imperial appearing in legacy contexts (K factor, circular mils). Be fluent in both voltage-drop methods shown in Section 8.

2. Ohm's law and resistance

To find	Formula	Units
Voltage	$E = I \times R$	volts (V)
Current	$I = E \div R$	amperes (A)
Resistance	$R = E \div I$	ohms (Ω)
Conductor resistance	$R = \rho L \div A$	Ω (length m, area m^2)
Series total resistance	$R_T = R_1 + R_2 + R_3 \dots$	Ω
Parallel (two resistors)	$R_T = (R_1 \times R_2) \div (R_1 + R_2)$	Ω
Parallel (any number)	$1/R_T = 1/R_1 + 1/R_2 + \dots$	Ω
Equal resistors in parallel	$R_T = R \div n$	Ω



WORKED EXAMPLE 2.1

Question: Three heating elements of 24 Ω , 24 Ω and 12 Ω are connected in parallel across a 240 V supply. What is the total current drawn?

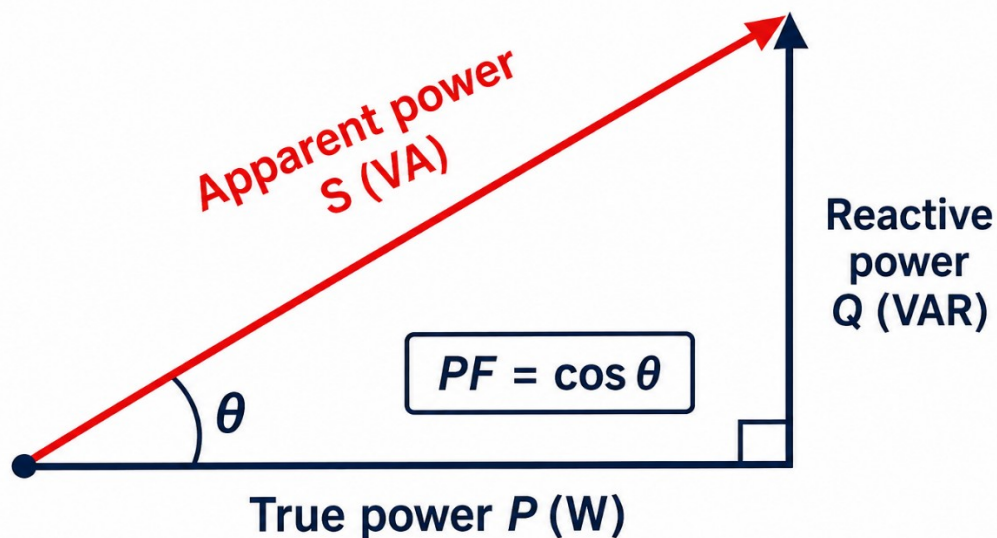
Step 1 — total resistance: $1/R_T = 1/24 + 1/24 + 1/12 = 1/24 + 1/24 + 2/24 = 4/24$, so $R_T = 6 \Omega$.

Step 2 — Ohm's law: $I = E \div R = 240 \div 6 = 40 \text{ A}$.

Exam check: parallel total resistance is always smaller than the smallest branch ($6 \Omega < 12 \Omega \checkmark$).

3. Power and energy

To find	Formula	Notes
Power (DC / resistive)	$P = E \times I$	watts
Power from current and R	$P = I^2 \times R$	the "I-squared-R" losses
Power from voltage and R	$P = E^2 \div R$	heaters, elements
AC true power (1 ϕ)	$P = E \times I \times PF$	watts (W)
Apparent power	$S = E \times I$	volt-amperes (VA)
Reactive power	$Q = E \times I \times \sin\theta$	VAR
Power factor	$PF = P \div S = \cos\theta$	dimensionless, 0–1
Energy	$W = P \times t$	kWh when P in kW, t in hours
Efficiency	$\eta = P_{out} \div P_{in} \times 100\%$	output over input, always < 100%



WORKED EXAMPLE 3.1

Question: A single-phase load draws 12 A at 240 V with a power factor of 0.85. Find the true power and the apparent power.

Apparent power: $S = 240 \times 12 = 2\,880 \text{ VA} = 2.88 \text{ kVA}$.

True power: $P = 240 \times 12 \times 0.85 = 2\,448 \text{ W} \approx 2.45 \text{ kW}$.

Exam check: true power is always $\leq$ apparent power. If your W exceeds your VA, you inverted the power factor.

4. AC fundamentals

Quantity	Formula	Notes
Period	$T = 1 \div f$	seconds; at 60 Hz, $T \approx 16.7$ ms
Peak from RMS	$E_{pk} = E_{RMS} \times 1.414$	120 V RMS $\rightarrow \approx 170$ V peak
RMS from peak	$E_{RMS} = E_{pk} \times 0.707$	meters read RMS
Inductive reactance	$X_L = 2\pi fL$	Ω ; increases with frequency
Capacitive reactance	$X_C = 1 \div (2\pi fC)$	Ω ; decreases with frequency
Impedance (series R and X)	$Z = \sqrt{R^2 + X^2}$	Ω
Circuit current	$I = E \div Z$	Ohm's law with impedance
Resonant frequency	$f_r = 1 \div (2\pi\sqrt{LC})$	$X_L = X_C$ at resonance

KEY POINT

In a purely inductive circuit current lags voltage by 90° ; in a purely capacitive circuit current leads by 90° . Remember ELI the ICE man: in an inductor (L), E leads I; in a capacitor (C), I leads E.

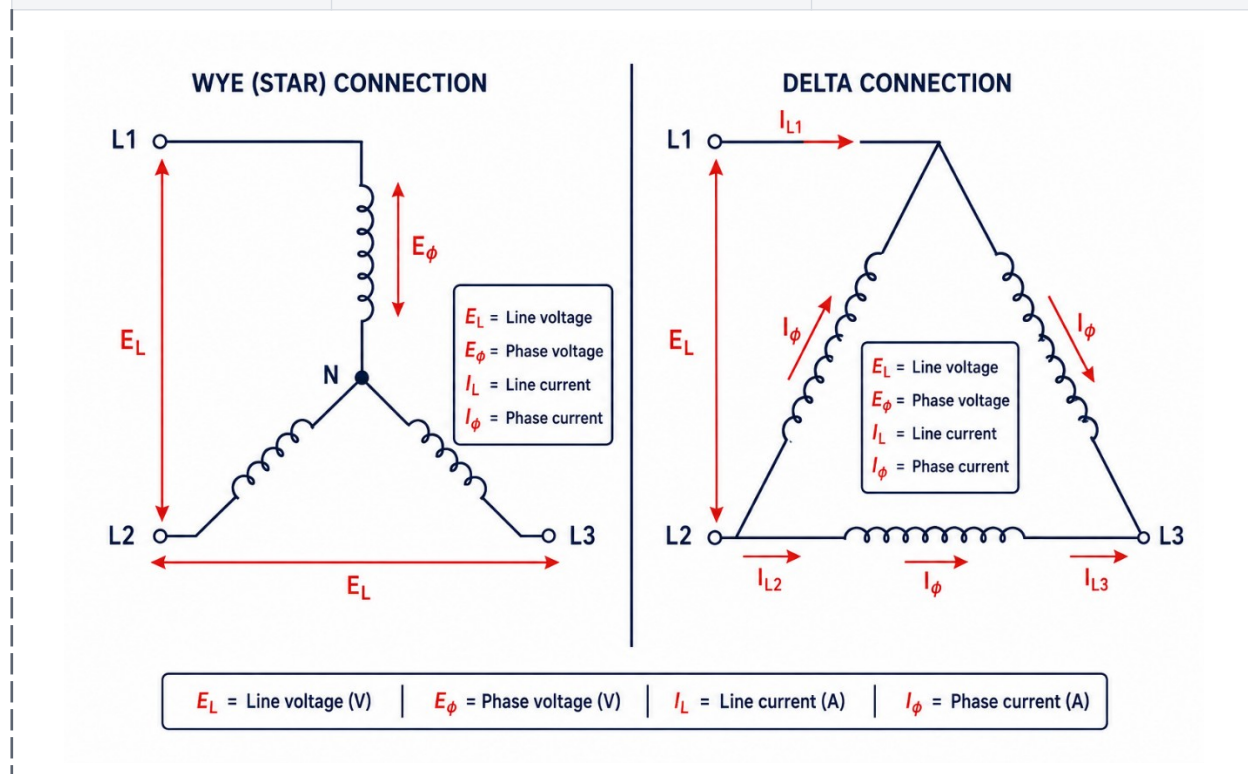
5. Single-phase and three-phase power

5.1 Formula set

System	Power formula	Current formula
Single-phase, resistive	$P = E \times I$	$I = P \div E$
Single-phase, with PF	$P = E \times I \times PF$	$I = P \div (E \times PF)$
Three-phase, true power	$P = \sqrt{3} \times E_L \times I_L \times PF$	$I_L = P \div (\sqrt{3} \times E_L \times PF)$
Three-phase, apparent power	$S = \sqrt{3} \times E_L \times I_L$	$I_L = S \div (\sqrt{3} \times E_L)$

5.2 Wye and delta relationships

Connection	Voltage relationship	Current relationship
Wye (star)	$E_L = E_\phi \times \sqrt{3}$	$I_L = I_\phi$
Delta	$E_L = E_\phi$	$I_L = I_\phi \times \sqrt{3}$



WORKED EXAMPLE 5.1

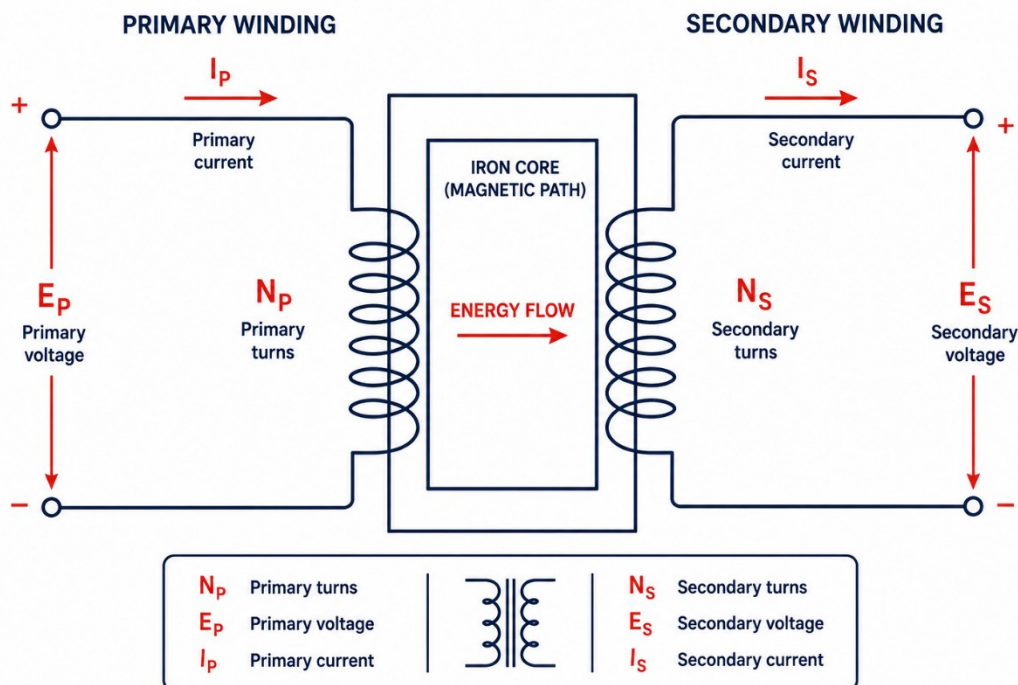
Question: A balanced three-phase heater is rated 15 kW at 600 V. What line current does it draw? (PF = 1 for resistive load.)

Solution: $I_L = P \div (\sqrt{3} \times E_L \times PF) = 15\,000 \div (1.732 \times 600 \times 1) = 15\,000 \div 1\,039 = 14.4\text{ A}$.

Exam check: for the same power, 600 V equipment draws roughly one-third the current of 208 V equipment. If your 600 V answer looks large, re-check.

6. Transformer formulas

To find	Formula	Notes
Turns / voltage ratio	$E_P \div E_S = N_P \div N_S$	voltage follows turns
Current ratio (inverse)	$I_P \div I_S = N_S \div N_P$	current opposes turns
Ideal power balance	$E_P \times I_P = E_S \times I_S$	kVA in = kVA out (ideal)
kVA rating (1 ϕ)	$kVA = E \times I \div 1000$	either winding
kVA rating (3 ϕ)	$kVA = \sqrt{3} \times EL \times IL \div 1000$	line values
Full-load current (1 ϕ)	$I = kVA \times 1000 \div E$	per winding voltage
Full-load current (3 ϕ)	$I = kVA \times 1000 \div (\sqrt{3} \times EL)$	line current
Efficiency	$\eta = P_{out} \div (P_{out} + \text{losses})$	copper + iron losses



WORKED EXAMPLE 6.1

Question: A dry-type transformer is rated 45 kVA, 600 V primary, 120/208 V three-phase secondary. Find the rated secondary line current.

Solution: $I = kVA \times 1000 \div (\sqrt{3} \times EL) = 45\,000 \div (1.732 \times 208) = 45\,000 \div 360 = 125\text{ A}$.

Memory hook: 45 kVA at 208 V $\rightarrow$ 125 A is a value worth memorizing outright; it appears constantly in distribution questions.

7. Motor formulas

To find	Formula	Notes
Synchronous speed	$nS = 120f \div P$	r/min; P = number of poles
Slip	$s = (nS - nR) \div nS \times 100\%$	typically 2–5% at full load
Rotor (shaft) speed	$nR = nS \times (1 - s)$	r/min
Output power	$P_{out} = hp \times 746$	watts
Input power (3 ϕ)	$P_{in} = \sqrt{3} \times EL \times IL \times PF$	watts
Motor efficiency	$\eta = (hp \times 746) \div P_{in}$	output over input
Full-load current (3 ϕ)	$I = hp \times 746 \div (\sqrt{3} \times E \times \eta \times PF)$	calculated; tables govern sizing
Poles	Synchronous speed at 60 Hz	
2	3 600 r/min	
4	1 800 r/min	
6	1 200 r/min	
8	900 r/min	

KEY POINT

For conductor and protection sizing on the exam, motor full-load current comes from the applicable CEC tables based on horsepower and voltage, not from nameplate or calculation. The calculated value is for theory questions; the table value is for installation questions. Know which one a question is asking for.

WORKED EXAMPLE 7.1

Question: A 4-pole, three-phase induction motor operates at 60 Hz with 3% slip. Find the shaft speed.

Step 1: $nS = 120 \times 60 \div 4 = 1\,800$ r/min.

Step 2: $nR = 1\,800 \times (1 - 0.03) = 1\,746$ r/min.

8. Voltage drop

Two accepted methods; be fluent in both. Under the CEC, the general limits are 3% for a branch circuit or feeder alone and 5% total from supply point to point of utilization. Always confirm limits and conductor data against the current edition of the code in force in your province.

8.1 Metric (resistivity) method

To find	Formula	Notes
Voltage drop (1 ϕ)	$VD = (2 \times \rho \times L \times I) \div A$	L one-way length (m), A in m ²
Voltage drop (3 ϕ)	$VD = (\sqrt{3} \times \rho \times L \times I) \div A$	replace 2 with $\sqrt{3}$
Percentage drop	$\%VD = VD \div E_{sys} \times 100$	compare to 3% / 5% limits

8.2 Imperial (K-factor) method

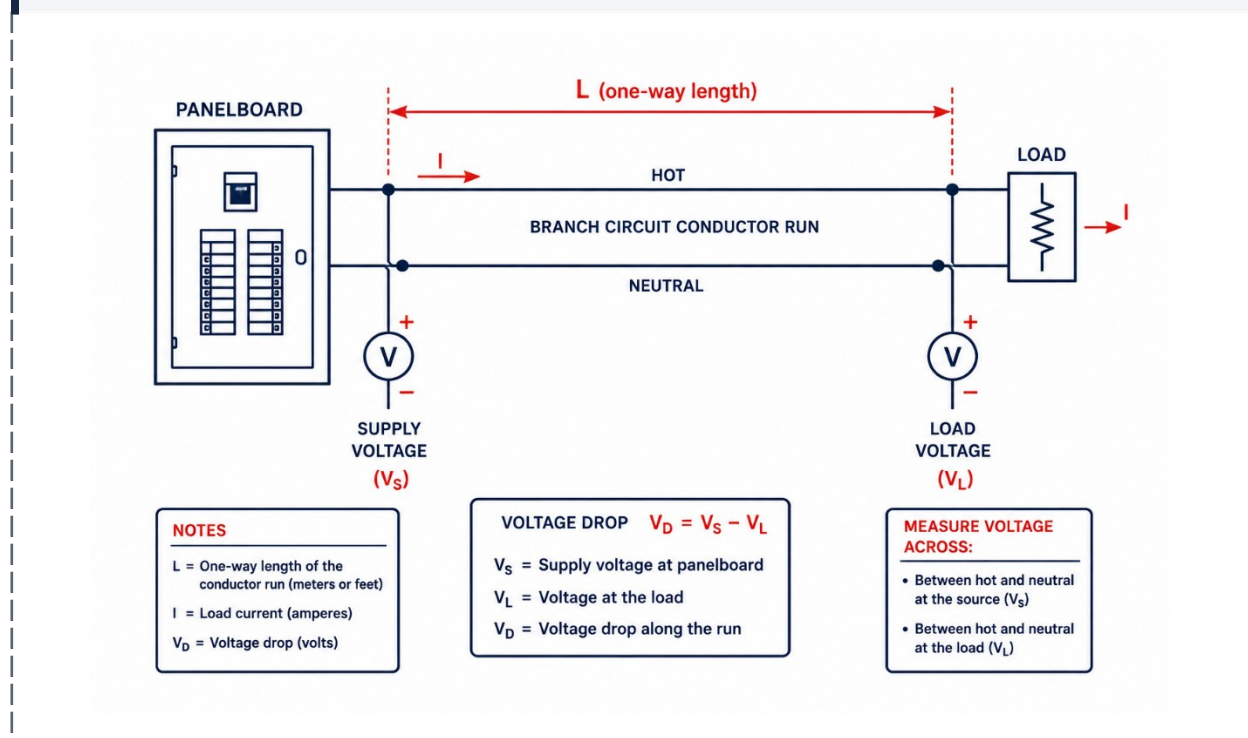
To find	Formula	Notes
Voltage drop (1 ϕ)	$VD = (2 \times K \times I \times L) \div CM$	L in feet, CM = circular mils
Voltage drop (3 ϕ)	$VD = (1.732 \times K \times I \times L) \div CM$	K $\approx$ 12.9 Cu, 21.2 Al
Minimum conductor size	$CM = (2 \times K \times I \times L) \div VD_{max}$	solve for CM, round up

WORKED EXAMPLE 8.1

Question: A 120 V single-phase load draws 16 A at the end of a 40 m run of copper conductor with a cross-section of 3.31 mm² (#12 AWG). Estimate the voltage drop. ($\rho = 1.72 \times 10^{-8} \Omega \cdot m$.)

Solution: $VD = (2 \times 1.72 \times 10^{-8} \times 40 \times 16) \div (3.31 \times 10^{-6}) = 2.20 \times 10^{-5} \div 3.31 \times 10^{-6} \approx 6.7 \text{ V}$.

Percentage: $6.7 \div 120 \times 100 \approx 5.5\%$ — over the limit; a larger conductor is required.



9. Unit conversions

From	To	Multiply by
horsepower (hp)	watts (W)	746
kilowatts (kW)	watts (W)	1 000
watts (W)	BTU/h	3.412
inches (in)	millimetres (mm)	25.4
feet (ft)	metres (m)	0.3048
metres (m)	feet (ft)	3.281
circular mils (cmil)	mm ²	0.000 506 7
mm ²	circular mils (cmil)	1 973.5
°C	°F	$(^{\circ}\text{C} \times 9/5) + 32$
kVA	kW	multiply by power factor

10. Electrical symbols

The exam expects instant recognition of standard drawing symbols. The table lists the symbols most frequently tested; the full chart belongs on your wall during study.

Symbol	Meaning
Circle with two parallel lines entering	Duplex receptacle
Circle with "GFCI"	Ground-fault circuit-interrupter receptacle
Circle with "WP"	Weatherproof device
S, S3, S4	Single-pole, three-way, four-way switch
Circle with four spokes	Luminaire (surface)
Circle inside square	Recessed luminaire
Zigzag line	Resistor / heating element
Coil arcs with two parallel bars	Transformer
Circle with "M"	Motor
Two parallel lines, one long one short	Battery / DC source
Triangle pointing to a line	Ground (earth) connection
Rectangle with diagonal	Panelboard
Dashed line	Wiring concealed in ceiling or wall
Line with perpendicular ticks	Number of conductors in run

STANDARD CANADIAN ELECTRICAL DRAWING SYMBOLS

1. RECEPTACLES (OUTLETS)							
Single Receptacle	Single Receptacle, Ground	Duplex Receptacle, Ground	Duplex Receptacle, Ground, Weatherproof	240 V Receptacle	Twist-Lock Receptacle	Floor Receptacle	Dedicated Receptacle
2. SWITCHES							
Single-Pole Switch	Single-Pole Three-Way Switch	Single-Pole Four-Way Switch	Double-Pole Switch	Key Switch	Push Button	Dimmer Switch	Timer Switch
3. LUMINAIRES (LIGHTING)							
Ceiling Light	Wall Light	Fluorescent Light	Strip Light	Recessed Light	Emergency Light	Exit Light	Outdoor Light
4. DISTRIBUTION							
Panelboard	Sub-Panelboard	Disconnect Switch	Fused Disconnect	Circuit Breaker	Bus Bars	Neutral Bus	Grounding Electrode
5. MOTORS AND CONTROLS							
Motor	Three-Phase Motor	Contactor	Overload Relay	Start Push Button	Stop Push Button	Selector Switch	Pilot Light
6. WIRING AND MISCELLANEOUS							
Conductor	Neutral Conductor	Grounding Conductor	Crossing Conductors (No Connection)	Connected Conductors (Connection)	Junction Box	Pull Box	Conduit

Note: Symbols are in accordance with CSA Z99 and common industry practice in Canada.

11. Formula index

Final-week drill sheet: cover the right column and reproduce each formula from its name.

Name	Formula
Ohm's law	$E = I \times R$
Power (resistive)	$P = E \times I$
I-squared-R losses	$P = I^2 R$
True power, 1φ	$P = EI \times PF$
True power, 3φ	$P = \sqrt{3} \times EL \times IL \times PF$
Energy	$W = P \times t$
Efficiency	$\eta = P_{out} \div P_{in}$
Inductive reactance	$X_L = 2\pi fL$
Capacitive reactance	$X_C = 1 \div 2\pi fC$
Impedance	$Z = \sqrt{R^2 + X^2}$
Wye line voltage	$E_L = E_\phi \times \sqrt{3}$
Delta line current	$I_L = I_\phi \times \sqrt{3}$
Transformer ratio	$E_P/E_S = N_P/N_S = I_S/I_P$
3φ transformer current	$I = kVA \times 1000 \div (\sqrt{3} \times E_L)$
Synchronous speed	$n_S = 120f \div P$
Slip	$s = (n_S - n_R) \div n_S$
Voltage drop, 1φ metric	$VD = 2pLI \div A$
Voltage drop, 1φ imperial	$VD = 2KIL \div CM$

FINAL TIP

If you can rebuild this index from memory twice on different days, formulas are no longer your risk area. Move your remaining study time to code navigation and installation questions.